Answer Key: Class-Book Misses

Cici Cisneros · SAT Math · Revision Bank #19 to #29 · November 11, 2026

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Questions are from Cici's SAT class book. Numbers match her Revision Bank cards on the hub. "Cici put" is her first answer on the page, before she corrected it.

#	Topic	Answer	Cici put	Error pattern
#19	Standard form: the slope is -a/b	B $\frac{1}{2}$	A $-\frac{1}{2}$	Sign slips moving terms
#20	Slope from standard form with a constant	A -6	D 6	Sign slips moving terms
#21	Perpendicular lines in standard form	C $\frac{3}{5}$	A $-\frac{3}{5}$	Sign slips moving terms
#22	Perpendicular with k inside the slope	D $\frac{3}{4}$	C $\frac{1}{4}$	Formula recall
#23	Slope tied to the y-intercept	A $\frac{20}{7}$	no answer bubbled	Set-up / translation
#24	Slope from a change in x and a change in y	C 2	no answer bubbled	Formula recall
#25	Function notation: f(x + 3) - f(x - 1)	C 27	no answer bubbled	Set-up / translation
#26	Average speed from a quadratic model	C 19.6 meters per second	A 4.9 meters per second	Formula recall
#27	Average rate of change with a constant k	D 10	A 4	Formula recall
#28	Infinitely many solutions: the point for every r	B $(r,\ 9-2r)$	C $(r,\ 18-2r)$	Arithmetic slip
#29	Tickets: two unknowns, two equations	120	100	Set-up / translation

#19 Standard form: the slope is -a/b

Sign slips moving terms

The line $ax + by = 18$ passes through the points $(2, 3)$ and $(6, 1)$. What is the value of $\frac{a}{b}$?

A $-\frac{1}{2}$ CICI B $\frac{1}{2}$ KEY C $-\frac{1}{4}$ D $\frac{1}{4}$

Your slope was right:

$$\frac{1-3}{6-2} = \frac{-2}{4} = -\frac{1}{2}$$

Your first move on the equation was right too: $y = \frac{-ax + 18}{b}$. Now split that fraction so the slope shows:

$$y = -\frac{a}{b}x + \frac{18}{b}$$

The slope is $-\frac{a}{b}$, minus sign included. Set that equal to the slope you found:

$$-\frac{a}{b} = -\frac{1}{2} \Rightarrow \frac{a}{b} = \frac{1}{2}$$

The miss was the very last step: writing $\frac{a}{b} = -\frac{1}{2}$ treats the slope as $\frac{a}{b}$ and loses the minus. Second way: plug in both points. $2a + 3b = 18$ and $6a + b = 18$ give $a = \frac{9}{4}$ and $b = \frac{9}{2}$, so $\frac{a}{b} = \frac{1}{2}$.

Fix: For ax + by = c the slope is -a/b. Write 'slope = -a/b' on the page before you set it equal to anything.

SAT class book, practice #1 (lines in standard form)

#20 Slope from standard form with a constant

Sign slips moving terms

When the equation $px - 3y = 12$ is rewritten in the form $y = mx + b$, the slope is -2. What is the value of p?

A -6 KEY B -2 C 2 **D** 6 CICI

Your first line was right:

$$y = \frac{-px + 12}{-3}$$

Now split it and let the -3 underneath act on each piece:

$$y = \frac{-p}{-3}x + \frac{12}{-3} = \frac{p}{3}x - 4$$

A negative over a negative is positive, so the slope is $\frac{p}{3}$. Set it equal to -2:

$$\frac{p}{3} = -2 \Rightarrow p = -6$$

Answer D, 6, comes from reading the slope as $-\frac{p}{3}$: one minus sign got used and the other got left behind. Check by plugging back in: $-6x - 3y = 12$ becomes $y = -2x - 4$, slope -2.

Fix: After dividing by a negative, rewrite the whole line with every sign shown, then read the slope. Plug your p back in to check.

SAT class book, practice #2 (lines in standard form)

#21 Perpendicular lines in standard form

Sign slips moving terms

The line $ax + by = 18$ is perpendicular to the line $5x - 3y = 12$. What is the value of $\frac{a}{b}$?

A $-\frac{3}{5}$ CICI B $-\frac{5}{3}$ **C** $\frac{3}{5}$ KEY D $\frac{5}{3}$

You rewrote the second line correctly: $y = \frac{5}{3}x - 4$, slope $\frac{5}{3}$. Perpendicular means flip and negate, so the first line needs slope $-\frac{3}{5}$. The slip was in the first line. Moving ax across the equals sign makes it negative:

$$by = -ax + 18$$

$$y = -\frac{a}{b}x + \frac{18}{b}$$

On paper it became $y = \frac{a}{b}x + \frac{18}{b}$, with the minus gone. Keep the sign and finish:

$$-\frac{a}{b} = -\frac{3}{5} \Rightarrow \frac{a}{b} = \frac{3}{5}$$

Answer A, $-\frac{3}{5}$, is the slope of the line. The question asks for $\frac{a}{b}$, which is the slope with its sign flipped.

Fix: Moving ax across gives by = -ax + 18, so the slope is -a/b. Then a/b is the slope with its sign flipped.

SAT class book, practice #4 (lines in standard form)

#22 Perpendicular with k inside the slope

Formula recall

Two lines are given by $y = (2k - 1)x + 5$ and $4x + 2y = 7$. For what value of k are the lines perpendicular?

A $-\frac{1}{4}$ B 0 C $\frac{1}{4}$ CICI D $\frac{3}{4}$ KEY

Your rewrite of the second line was right: $y = -2x + \frac{7}{2}$, slope -2. The first line is already in slope-intercept form, so its slope is the whole bracket, $2k - 1$. No need to expand it. Perpendicular: flip -2 and negate it, which gives $+\frac{1}{2}$.

$$2k - 1 = \frac{1}{2} \Rightarrow 2k = \frac{3}{2} \Rightarrow k = \frac{3}{4}$$

Answer C, $\frac{1}{4}$, is the trap. It comes out of $2k - 1 = -\frac{1}{2}$ (flipped but not negated) and also out of $2k = \frac{1}{2}$ (the -1 left behind). Check: $k = \frac{3}{4}$ makes the first slope $\frac{1}{2}$, and $\frac{1}{2} \cdot (-2) = -1$.

Fix: Three separate lines on the page: slope 1, slope 2, perpendicular slope. Then check that the two slopes multiply to -1.

SAT class book, practice #6, page 18 (lines in standard form)

#23 Slope tied to the y-intercept

Set-up / translation

The slope of a line is 3 times its y-intercept. If the line passes through $(2, 20)$, what is the y-intercept?

A $\frac{20}{7}$ KEY B 4 C $\frac{30}{7}$ D 5

Cici put: **no answer bubbled**

Say it, then write it. 'The slope is 3 times the y-intercept' does not say the slope is 3. Call the y-intercept b; then the slope is $3b$:

$$y = 3bx + b$$

Plug in $(2, 20)$:

$$20 = 3b(2) + b = 7b \Rightarrow b = \frac{20}{7}$$

On paper you used a slope of 3 and got $y = 3x + 14$. That line does pass through $(2, 20)$, but its slope (3) is not 3 times its intercept (14), and 14 is not one of the choices. When your answer is not on the list, go back to the sentence before you go back to the algebra.

Fix: Name the unknown first (b = y-intercept), then translate the sentence: slope = 3b. An answer that is not a choice means reread the question.

SAT class book, practice #5 (slope and intercept)

#24 Slope from a change in x and a change in y

Formula recall

The equation of a line is written in slope-intercept form $y = mx + b$. If increasing x by 4 increases y by 10, and the line passes through $(2, 7)$, what is the value of b?

A -1 B 0 **C 2 KEY** D 5

Cici put: **no answer bubbled**

Slope is the change in y over the change in x, rise over run:

$$m = \frac{10}{4} = \frac{5}{2}$$

On paper it went in upside down, as $\frac{4}{10} = \frac{2}{5}$. Now use the point $(2, 7)$:

$$7 = \frac{5}{2}(2) + b = 5 + b \;\Rightarrow\; b = 2$$

Two things to notice. First, y grows faster than x here (10 for every 4), so the slope has to be bigger than 1; $\frac{2}{5}$ could not be right. Second, when you distributed $\frac{2}{5}(x - 2)$, the $-\frac{4}{5}$ turned into $+\frac{4}{5}$, which gave 7.8. That is not a choice either, and that is the signal to recheck the slope.

Fix: Slope = change in y over change in x. Write 'rise / run' above the fraction before you fill it in.

SAT class book, practice #6 (slope and intercept)

#25 Function notation: f(x + 3) - f(x - 1)

Set-up / translation

A function f is linear and satisfies $f(x + 3) - f(x - 1) = 20$ for all real x. If $f(2) = 7$, what is $f(6)$?

A 17 B 21 **C 27 KEY** D 31

Cici put: **no answer bubbled**

Plugging in numbers was a good instinct. The fastest choice is the x that makes the two inputs the ones you care about, 6 and 2. That is $x = 3$:

$$f(3 + 3) - f(3 - 1) = f(6) - f(2) = 20$$

Since $f(2) = 7$:

$$f(6) = 20 + 7 = 27$$

The wrong turn on paper was $f(5) - f(1) = 20$ becoming $f(4) = 20$. Inputs cannot be subtracted like that: $f(5)$ and $f(1)$ are two separate outputs, and their difference is just the number 20. What that line really says is 'inputs 4 apart, outputs 20 apart', which is a slope of $\frac{20}{4} = 5$. Second way, using the slope: from $x = 2$ to $x = 6$ is 4 steps of 5, so $f(6) = 7 + 20 = 27$.

Fix: f(5) - f(1) is two outputs subtracted; it is never f(4). Pick the x that gives the inputs you want: x = 3 makes it f(6) - f(2) = 20.

SAT class book, practice #7 (slope and intercept)

#26 Average speed from a quadratic model

Formula recall

An object is dropped from a cliff. The distance, in meters, that it has fallen after t seconds is modeled by the function $d(t) = 4.9t^2$. What is the average speed (average rate of change) over the interval from $t = 1$ to $t = 3$?

A 4.9 meters per second CICI **B** 9.8 meters per second **C** 19.6 meters per second KEY

D 39.2 meters per second

Average rate of change is a slope between two points on the curve: change in output over change in input. Evaluate each endpoint on its own line, squaring first:

$$d(3) = 4.9(9) = 44.1$$
$$d(1) = 4.9(1) = 4.9$$

Then one fraction:

$$\frac{d(3) - d(1)}{3 - 1} = \frac{39.2}{2} = 19.6$$

Every wrong choice here is a step left out. 39.2 is the top of the fraction without dividing by 2. 4.9 is what comes out if the t values never get squared: $\frac{14.7-4.9}{2}$. 9.8 is that same top without dividing.

Fix: Write (d(3) - d(1)) / (3 - 1) first, then fill in each piece on its own line. Square before you multiply.

SAT class book, practice (average rate of change)

#27 Average rate of change with a constant k

Formula recall

The average rate of change of a function $f(x) = x^2 + kx$ from $x = 3$ to $x = 7$ is 20. What is the value of k?

A 4 CICI **B** 6 **C** 8 **D** 10 KEY

Same formula as any average rate of change, with the letter k riding along. Evaluate each endpoint on its own line:

$$f(7) = 49 + 7k$$
$$f(3) = 9 + 3k$$

Subtract with the parentheses kept:

$$(49 + 7k) - (9 + 3k) = 40 + 4k$$

Divide by the change in x, which is $7 - 3 = 4$, and set it equal to 20:

$$\frac{40 + 4k}{4} = 10 + k = 20 \;\Rightarrow\; k = 10$$

Check it a second way: with $k = 10$, $f(7) = 119$ and $f(3) = 39$, and $\frac{119-39}{4} = 20$. The same check on $k = 4$ gives $\frac{77-21}{4} = 14$, not 20, so plugging the answer back in would have caught it.

Fix: Average rate of change = (f(7) - f(3)) / (7 - 3). Keep k in both endpoints, then plug your answer back in to check.

SAT class book, practice (average rate of change)

#28 Infinitely many solutions: the point for every r

Arithmetic slip

Consider the system of equations:

$$4x + 2y = 18$$
$$8x + 4y = 36$$

For each real number r, which of the following points lies on the graph of both equations in the xy-plane?

A $(r,\ 2r - 9)$ B $(r,\ 9 - 2r)$ KEY C $(r,\ 18 - 2r)$ CICI D $(9 - 2r,\ r)$

The second equation is the first one doubled, so the two graphs are the same line and every point on that line works. The question is really asking: if $x = r$, what is y? Solve the first equation for y, dividing every term by 2:

$$2y = 18 - 4x \ \Rightarrow\ y = 9 - 2x$$

So the point is $(r,\ 9 - 2r)$. On paper the factoring went $2(4x + y = 18)$, which matches neither equation: pulling a 2 out of $8x + 4y = 36$ gives $2(4x + 2y) = 36$. That led to $y = 18 - 4r$, which is not on the list, and C was the closest-looking choice. Fast check for any 'for each r' question: try $r = 0$. The first equation gives $y = 9$, so the point has to be $(0, 9)$, and only B gives that.

Fix: Divide every term by the same number. Then test the choices with r = 0: the right one has to land on the line.

SAT class book, practice #1 (infinitely many solutions)

#29 Tickets: two unknowns, two equations

Set-up / translation

Tickets for a movie cost $8.00 for adults and $5.00 for children. A total of 200 tickets were sold, and the total revenue was $1,360. How many adult tickets were sold?

Answer: 120 · Cici put: 100

Two different things are being counted, so they need two different letters. Let a be the adult tickets and c the child tickets. One equation counts tickets, the other counts dollars:

$$a + c = 200$$
$$8a + 5c = 1360$$

From the first, $c = 200 - a$. Substitute:

$$8a + 5(200 - a) = 1360$$
$$3a + 1000 = 1360 \ \Rightarrow\ a = 120$$

Check both sentences: $120 + 80 = 200$ tickets, and $8(120) + 5(80) = 960 + 400 = 1360$ dollars. On paper both kinds of ticket were called x, which made $8x + 5x = 13x$ and mixed the ticket total with the dollar total. One letter can only stand for one thing.

Fix: Two unknowns, two letters, two equations: one counts items, one counts money. Write what each letter stands for before any algebra.

SAT class book, practice #8 (systems word problems)